FORGE

User Stories (Revised)

This document contains 58 user stories for FORGE, grouped by feature area. Every story names one actor from the glossary, has a MoSCoW priority, traces to the functional or non-functional requirements, and has acceptance criteria written as testable Given/When/Then statements. System-quality requirements are in Section 5 and external dependencies are in Section 6. Values that earlier drafts left undefined are now set in Section 3; they are proposals for the team to confirm.

1. Actor Glossary

Only these actors are used in the stories. Earlier terms map as follows: Regular Feed User, User and Project User become Registered User; Contributor / Team Member becomes Contributor; Project Owner / Creator becomes Project Owner; Project Admin / Manager becomes Project Manager.

Actor	Definition
Visitor	Anyone who is not logged in. Can browse and view public projects and register.
Registered User	Anyone with a FORGE account who is logged in. Base actor for profile, direct messages, public posts and discovery.
Project Owner	A Registered User with the Owner role on a project. Usually the creator. Has every right in the project.
Project Manager	A project member with the Admin or Owner role. Used when a story applies to both.
Contributor	A project member with the Contributor role or higher (Admin, Owner). Can create and edit project work.
Project Member	A project member with any role, including Viewer.
Content Moderator	A Registered User granted the moderator role by a Platform Admin. Reviews reported public posts.
Platform Admin	A user with platform-wide administrative rights, independent of any project.

System and external actors (AI Project Copilot, GitHub, LLM API provider, hosting providers) are not written as user stories. See Section 6.

2. Project Role-Permission Matrix

Roles are cumulative: Owner includes all Admin rights, Admin includes all Contributor rights, Contributor includes all Viewer rights. A dash means the role cannot perform the action; the server must reject it with 403. A Viewer cannot create tasks.

Action	Owner	Admin	Contributor	Viewer
View project content, Activity Feed, dashboard	Yes	Yes	Yes	Yes
Read channels and task comments	Yes	Yes	Yes	Yes
Use Copilot and Catch Me Up	Yes	Yes	Yes	Yes
Post and reply in channels, mention members	Yes	Yes	Yes	-
Create, edit, assign tasks; set due dates; comment	Yes	Yes	Yes	-
Change task status, drag on Kanban board	Yes	Yes	Yes	-
Attach tasks to milestones	Yes	Yes	Yes	-

Action	Owner	Admin	Contributor	Viewer
Record and edit decisions	Yes	Yes	Yes	-
Upload files, write documentation pages	Yes	Yes	Yes	-
Create milestones	Yes	Yes	-	-
View project health	Yes	Yes	-	-
Invite and remove members, approve join requests	Yes	Yes	-	-
Assign roles	Yes	Yes (not Owner)	-	-
Disconnect or re-authorize GitHub	Yes	Yes	-	-
Authorize GitHub and connect a repository	Yes	-	-	-
Mark project complete or reopen it	Yes	-	-	-
Leave project	Yes (not sole Owner)	Yes	Yes	Yes

3. Defined Values Used in Acceptance Criteria

Proposed values. The team should confirm or change them before implementation; the stories and the quality table in Section 5 use them.

Item	Value
Password	At least 8 characters, stored as a salted hash
Session (JWT) expiry	24 hours
Failed-login lock	5 consecutive failures lock login for 15 minutes
Invitation expiry	7 days
Maximum project file size	10 MB
Profile photo	JPEG or PNG, up to 2 MB; bio up to 500 characters; up to 10 skill tags
Public post	Text up to 500 characters; one optional JPEG, PNG or WebP image up to 5 MB
Rate limits (per user)	10 posts per hour; 30 votes per minute
Trending score	(upvotes - downvotes) / (age in hours + 2)^1.5, recalculated every 5 minutes
Total likes (profile)	Total upvotes received across a user's public posts
Deadline notification	Sent when a task that is not Done is less than 24 hours from its due date
Overdue / blocked task	Overdue: past due date and not Done. Blocked: carries the label "Blocked"
Activity level	Low: fewer than 5 feed events in 7 days; Medium: 5 to 20; High: more than 20
Since last login (Catch Me Up)	Time of the user's previous successful login
Copilot limits	Question up to 500 characters; model input up to 8,000 tokens; loading indicator after 2 seconds
Collaborator search	Match any selected tag, case-insensitive, ordered by number of matching tags
Portfolio statistics	Tasks completed, decisions recorded, messages posted, per completed project

Item	Value
Public project page	Name, description, domain, status, member names. No channels, messages, tasks or files

4. Functional User Stories

4.1 Accounts and Profiles

US-01 **Register an account**	Actor: Visitor User Story: As a Visitor, I want to register with my email address and a password, so that I can access protected FORGE features. Priority: Must Traceability: FR-1; NFR-3a
Acceptance Criteria	• **Given** I am on the sign-up page, **when** I submit a valid, unused email and a password of at least 8 characters, **then** my account is created, my password is stored only as a salted hash, and I am logged in. • **Given** an account already exists for the email, **when** I submit the sign-up form, **then** I see an error message and no second account is created.

US-02 **Log in**	Actor: Registered User User Story: As a Registered User, I want to log in with my email and password, so that I get a time-limited session for my projects and messages. Priority: Must Traceability: FR-1, FR-28; NFR-3b, NFR-3c
Acceptance Criteria	• **Given** I enter valid credentials, **when** I submit the login form, **then** I receive a JWT session that expires after 24 hours. • **Given** I have entered wrong credentials 5 times in a row, **when** I try to log in again, **then** login is refused for 15 minutes and I see a message saying so. • **Given** my session has expired, **when** I request a protected resource, **then** the request is rejected with 401 and I am sent to the login page.

US-03 **Log out**	Actor: Registered User User Story: As a Registered User, I want to log out, so that nobody else can use my session on a shared device. Priority: Must Traceability: FR-28; NFR-22
Acceptance Criteria	• **Given** I am logged in, **when** I choose Log out, **then** my token is invalidated and any later request using it returns 401.

US-04 **Reset a forgotten password**	Actor: Visitor User Story: As a Visitor, I want to reset my password by email, so that I can regain access to my account. Priority: Should Traceability: New (review); FR-1
Acceptance Criteria	• **Given** I submit a registered email on the reset page, **when** the request is processed, **then** a single-use reset link valid for 1 hour is emailed to me. • **Given** I submit an email that is not registered, **when** the request is processed, **then** I see the same confirmation message, so account existence is not revealed. • **Given** I open a valid reset link and enter a new valid password, **when** I submit, **then** the password is updated and all my existing sessions are invalidated.

US-05 **Edit my profile**	Actor: Registered User User Story: As a Registered User, I want to add and edit my photo, bio and skill tags, so that other users can understand my background and skills. Priority: Must Traceability: FR-2, FR-29
Acceptance Criteria	• **Given** I am on my profile settings, **when** I upload a JPEG or PNG photo up to 2 MB and a bio up to 500 characters and save, **then** the new photo and bio appear on my profile. • **Given** I have fewer than 10 skill tags, **when** I add, rename or remove a tag and save, **then** my profile shows the updated tag list. • **Given** I upload a photo above 2 MB or in an unsupported format, **when** I save, **then** I see a clear error and my previous photo is kept.

US-06 **View another user's profile**	Actor: Registered User User Story: As a Registered User, I want to view another user's profile, so that I can learn about their skills and public activity before collaborating. Priority: Should Traceability: New (review); FR-19, FR-26
Acceptance Criteria	• **Given** I am logged in, **when** I open another user's profile, **then** I see their photo, bio, skill tags, follower count and total likes. • **Given** that user has hidden fields via privacy settings, **when** I open their profile, **then** the hidden fields are not shown.

US-07 **Set privacy preferences**	Actor: Registered User User Story: As a Registered User, I want to control whether I appear in collaborator search and whether my portfolio is public, so that I decide how much of my information is discoverable. Priority: Could Traceability: New (review); FR-19, FR-21
Acceptance Criteria	• **Given** I have turned off "appear in collaborator search", **when** another user searches by one of my skill tags, **then** my profile is not returned. • **Given** I have turned off "public portfolio", **when** someone opens my portfolio link, **then** they see a "not available" page.

4.2 Projects and Membership

US-08 **Create a project**	Actor: Registered User User Story: As a Registered User, I want to create a project with a name, description, domain, visibility and starting members, so that I have a workspace where project work is organized. Priority: Must Traceability: FR-3, FR-18
Acceptance Criteria	• **Given** I am logged in, **when** I submit a name (up to 80 characters), a description, a domain from the fixed list and visibility Public or Private, **then** the project is created and I am its Owner. • **Given** I add existing users as starting members and give each a role, **when** I create the project, **then** each starting member is added with that role. • **Given** I leave the name empty, **when** I submit, **then** I see a validation error and no project is created.

US-09 **Assign project roles**	Actor: Project Manager User Story: As a Project Manager, I want to assign the Admin, Contributor or Viewer role to project members, so that each member has the right permissions. Priority: Must Traceability: FR-4; NFR-19
Acceptance Criteria	• **Given** I am an Admin or Owner of the project, **when** I change a member's role, **then** the new role applies to that member's next request, according to the permission matrix. • **Given** I am an Admin, **when** I try to change the Owner's role or grant Owner, **then** the action is refused. • **Given** I am a Contributor or Viewer, **when** I try to change a role, **then** the request is rejected with 403.

US-10 **Invite an existing user**	Actor: Project Manager User Story: As a Project Manager, I want to invite an existing FORGE user to my project with a chosen role, so that I can grow the team after the project is created. Priority: Should Traceability: FR-31
Acceptance Criteria	• **Given** I am an Admin or Owner, **when** I invite a registered user by email or username, **then** a pending invitation appears in that user's invitations list and expires after 7 days. • **Given** the user is already a member or already has a pending invitation, **when** I invite them again, **then** I see an error and no duplicate is created.

US-11 **Accept or decline an invitation**	Actor: Registered User User Story: As a Registered User, I want to accept or decline a project invitation, so that I control which projects I join. Priority: Should Traceability: New (review); FR-31
Acceptance Criteria	• **Given** I have a pending invitation, **when** I accept it, **then** I become a member with the invited role and the invitation is closed. • **Given** I have a pending invitation, **when** I decline it, **then** I am not added and the inviter can see the invitation was declined.

US-12 **Approve or reject join requests**	Actor: Project Manager User Story: As a Project Manager, I want to review join requests for my public project, so that I decide who joins. Priority: Should Traceability: New (review); FR-20
Acceptance Criteria	• **Given** a user has submitted a join request, **when** I approve it, **then** the user becomes a Contributor (or the role I choose) and can open the project. • **Given** a user has submitted a join request, **when** I reject it, **then** the user is not added and may not re-request for 7 days.

US-13 **Remove a member**	Actor: Project Manager User Story: As a Project Manager, I want to remove a member from my project, so that people who no longer take part lose access. Priority: Should Traceability: New (review); FR-4
Acceptance Criteria	• **Given** I am an Admin or Owner, **when** I remove a member, **then** that member can no longer open the project, and their past messages and tasks remain. • **Given** the member is the only Owner, **when** I try to remove them, **then** the action is refused with an explanation.

US-14 **Leave a project**	Actor: Project Member User Story: As a Project Member, I want to leave a project, so that I can step away from work I am no longer doing. Priority: Should Traceability: New (review); FR-4
Acceptance Criteria	• **Given** I am a member but not the only Owner, **when** I choose Leave project and confirm, **then** I lose access and my past contributions stay in the project. • **Given** I am the only Owner, **when** I choose Leave project, **then** I am asked to transfer ownership first.

US-15 **Mark a project complete**	Actor: Project Owner User Story: As a Project Owner, I want to mark my project as Completed, so that it appears as finished work in members' portfolios. Priority: Should Traceability: New (review); FR-21
Acceptance Criteria	• **Given** I am the Owner of an active project, **when** I mark it Completed, **then** its status changes to Completed and it becomes eligible for members' portfolios. • **Given** a project is Completed, **when** the Owner reopens it, **then** its status returns to Active.

4.3 Communication

US-16 **Send messages in project channels**	Actor: Contributor User Story: As a Contributor, I want to send messages in my project's text channels, so that my team can communicate in one organized place. Priority: Must Traceability: FR-5; NFR-1b
Acceptance Criteria	• **Given** I am a Contributor in a project with a #general channel, **when** I send a message, **then** every member currently viewing the channel sees it within 1 second and it is saved in the history. • **Given** I am a Viewer, **when** I try to send a message, **then** the request is rejected with 403, and I can still read the channel.

US-17 **Reply in a thread**	Actor: Contributor User Story: As a Contributor, I want to reply to a channel message in a thread, so that side discussions do not clutter the channel. Priority: Should Traceability: FR-5
Acceptance Criteria	• **Given** a channel message exists, **when** I post a reply to it, **then** the reply appears in that message's thread and the message shows its reply count.

US-18 **Mention a teammate**	Actor: Contributor User Story: As a Contributor, I want to mention a project member with @name, so that they notice messages meant for them. Priority: Must Traceability: FR-5, FR-10
Acceptance Criteria	• **Given** I type @ in a message, **when** I continue typing, **then** I see suggestions limited to current project members. • **Given** I send a message mentioning a member, **when** the message is delivered, **then** that member receives a mention notification linking to the message.

US-19 **Send direct messages**	Actor: Registered User User Story: As a Registered User, I want to send and receive direct messages with another user, so that I can talk privately outside project channels. Priority: Must Traceability: FR-6; NFR-1b
Acceptance Criteria	• **Given** I open another user's profile, **when** I choose Message, **then** a direct conversation opens, separate from all project channels. • **Given** I send a direct message, **when** it is delivered, **then** the recipient sees it within 1 second and only the two participants can read it.

US-20 **Comment on a task**	Actor: Contributor User Story: As a Contributor, I want to comment on a task, so that discussion about the work stays with the task. Priority: Should Traceability: New (review); FR-7
Acceptance Criteria	• **Given** I am viewing a task, **when** I post a comment, **then** it appears under the task with my name and a timestamp. • **Given** my comment mentions a member with @name, **when** I post it, **then** that member receives a mention notification.

4.4 Task Management

US-21 **Create a task**	Actor: Contributor User Story: As a Contributor, I want to create a task with a title, description and label, so that work is captured clearly. Priority: Must Traceability: FR-7a
Acceptance Criteria	• **Given** I am a Contributor, **when** I submit a title (required), an optional description and an optional label, **then** the task is created in the project with status To Do. • **Given** I am a Viewer, **when** I try to create a task, **then** the request is rejected with 403.

US-22 **Assign a task**	Actor: Contributor User Story: As a Contributor, I want to assign a task to a project member, so that responsibility is clear. Priority: Must Traceability: FR-7a, FR-10
Acceptance Criteria	• **Given** a task exists, **when** I choose an assignee from the current project members, **then** the task shows that assignee and the assignee receives an assignment notification. • **Given** the user is not a project member, **when** I try to assign them, **then** they cannot be selected and an API attempt is rejected.

US-23 **Set a task due date**	Actor: Contributor User Story: As a Contributor, I want to set a due date on a task, so that the team can see and track deadlines. Priority: Should Traceability: New (review); FR-10, FR-15
Acceptance Criteria	• **Given** a task exists, **when** I set a due date, **then** the date is shown on the task and in the dashboard's upcoming deadlines. • **Given** a task has a due date and is not Done, **when** the date is less than 24 hours away, **then** the assignee receives a deadline notification.

US-24 **Update task status**	Actor: Contributor User Story: As a Contributor, I want to change a task's status between To Do, In Progress and Done, so that the team sees the current state of work. Priority: Must Traceability: FR-7b, FR-32
Acceptance Criteria	• **Given** a task exists, **when** I change its status, **then** the new status is saved and other members see it within 3 seconds. • **Given** a task's status changes, **when** the change is saved, **then** it appears as an entry in the project Activity Feed.

US-25 **Create a milestone**	Actor: Project Manager User Story: As a Project Manager, I want to create a milestone with a title and optional target date, so that related work can be tracked as a larger goal. Priority: Must Traceability: FR-7c
Acceptance Criteria	• **Given** I am an Admin or Owner, **when** I submit a milestone title, **then** the milestone is created and listed in the project. • **Given** I am a Contributor or Viewer, **when** I try to create a milestone, **then** the request is rejected with 403.

US-26 **Group tasks under a milestone**	Actor: Contributor User Story: As a Contributor, I want to attach one or more tasks to a milestone, so that progress toward the goal is visible. Priority: Must Traceability: FR-7c
Acceptance Criteria	• **Given** a milestone and tasks exist, **when** I attach tasks to the milestone, **then** the milestone lists them and shows progress as completed tasks out of total tasks.

US-27 **View the Kanban board**	Actor: Project Member User Story: As a Project Member, I want to see tasks on a Kanban board with To Do, In Progress and Done columns, so that I can see work at a glance. Priority: Must Traceability: FR-8; NFR-5b
Acceptance Criteria	• **Given** the project has tasks, **when** I open the board, **then** each task appears in the column matching its status.

US-28 **Move tasks by dragging**	Actor: Contributor User Story: As a Contributor, I want to drag a task card to another column, so that I can update status quickly. Priority: Must Traceability: FR-8; NFR-5b, NFR-25
Acceptance Criteria	• **Given** I drag a card to another column and drop it, **when** the drop completes, **then** the card moves within 300 ms and the status is saved. • **Given** the server rejects the update, **when** the response arrives, **then** the card returns to its original column and I see an error. • **Given** another member changed the same task after I loaded the board, **when** I save my change, **then** my change is rejected with a conflict message and the latest version is shown, so nothing is silently overwritten.

4.5 Activity Feed and Notifications

US-29 **View the project Activity Feed**	Actor: Project Member User Story: As a Project Member, I want to see one chronological feed of project messages, task changes and GitHub events, so that important activity is in one place. Priority: Must Traceability: FR-9; NFR-1a
Acceptance Criteria	• **Given** a project has messages, task changes and connected GitHub events, **when** I open the Activity Feed, **then** they appear in one list, newest first, each showing type, actor and time. • **Given** the feed has up to 100 concurrent viewers, **when** it loads, **then** it loads within 3 seconds. • **Given** I am not a member of a private project, **when** I request its feed, **then** the request is rejected with 403.

US-30 **Receive in-app notifications**	Actor: Registered User User Story: As a Registered User, I want to receive in-app notifications for mentions, task assignments and approaching deadlines, so that I do not miss important project actions. Priority: Must Traceability: FR-10, FR-33; NFR-6
Acceptance Criteria	• **Given** I am mentioned, assigned a task, or have a task due within 24 hours, **when** the event occurs, **then** a notification appears in my notification list within 5 seconds, labelled Mention, Assignment or Deadline. • **Given** I open a notification, **when** I select it, **then** I am taken to the related message or task and it is marked as read.

4.6 GitHub Integration

US-31 **Authorize GitHub**	Actor: Project Owner User Story: As a Project Owner, I want to authorize FORGE to access my GitHub account through OAuth, so that FORGE can read my repositories. Priority: Must Traceability: FR-11
Acceptance Criteria	• **Given** I open project integration settings, **when** I choose Connect GitHub and approve access on GitHub, **then** I return to FORGE with authorization stored. • **Given** I deny access on GitHub, **when** I return to FORGE, **then** I see a message that the connection was not made and nothing is stored.

US-32 **Connect a repository**	Actor: Project Owner User Story: As a Project Owner, I want to connect one GitHub repository to my project, so that its development activity flows into the project. Priority: Must Traceability: FR-11, FR-35
Acceptance Criteria	• **Given** I have authorized GitHub, **when** I pick a repository from my list, **then** the repository is connected and the project shows the connection as Active.

US-33 **Manage the GitHub connection**	Actor: Project Manager User Story: As a Project Manager, I want to view, disconnect or re-authorize the GitHub connection, so that I can fix or end the integration. Priority: Should Traceability: FR-35
Acceptance Criteria	• **Given** a repository is connected, **when** I open integration settings, **then** I see the repository name and connection status. • **Given** I disconnect the repository, **when** I confirm, **then** no further GitHub events are added to the feed and earlier events remain. • **Given** the authorization has expired, **when** I choose Re-authorize, **then** the connection returns to Active after I approve on GitHub.

US-34 **See GitHub events in the feed**	Actor: Project Member User Story: As a Project Member, I want to see commits, issues and pull requests from the connected repository in the Activity Feed, so that development activity stays tied to project context. Priority: Must Traceability: FR-12; NFR-11a, NFR-11b
Acceptance Criteria	• **Given** a repository is connected, **when** a commit is pushed, or an issue or pull request is opened or updated, **then** a feed entry showing event type, title, author and a link to GitHub appears within 60 seconds.

4.7 Decisions, Files, Documentation and Project Insight

US-35 **Record a decision**	Actor: Contributor User Story: As a Contributor, I want to record a decision with a title, description, participants, date and a link to a related task or discussion, so that the reasons behind project choices are kept. Priority: Must Traceability: FR-13
Acceptance Criteria	• **Given** I am a Contributor, **when** I submit a title, description, at least one participant from the project members and a date, **then** the decision is saved. • **Given** I select a related task or channel thread, **when** I save, **then** the decision shows a working link to it. • **Given** a required field is missing, **when** I save, **then** I see a validation error and nothing is saved.

US-36 **View and update decisions**	Actor: Contributor User Story: As a Contributor, I want to view and edit recorded decisions, so that decision records stay accurate. Priority: Must Traceability: FR-34
Acceptance Criteria	• **Given** decisions exist, **when** a Project Member opens the decision list, **then** they see all decisions with title and date. • **Given** I am a Contributor, **when** I edit a decision and save, **then** the updated details are shown.

US-37 **Upload a project file**	Actor: Contributor User Story: As a Contributor, I want to upload a file to my project, so that project resources are stored with the project. Priority: Should Traceability: FR-14a; NFR-8
Acceptance Criteria	• **Given** I select a file of 10 MB or less, **when** I upload it, **then** it is listed in the project's files and members can download it. • **Given** I select a file larger than 10 MB, **when** I upload it, **then** it is rejected with a message stating the 10 MB limit.

US-38 **Write a documentation page**	Actor: Contributor User Story: As a Contributor, I want to write a Markdown documentation page of up to 10,000 characters, so that project knowledge is documented simply. Priority: Should Traceability: FR-14b
Acceptance Criteria	• **Given** I write Markdown text and save, **when** the page is saved, **then** it is listed under the project's documentation and displays as formatted text. • **Given** I am a Project Member, **when** I open a documentation page, **then** I can read it.

US-39 **View the project dashboard**	Actor: Project Member User Story: As a Project Member, I want to see milestone progress, open task counts and upcoming deadlines, so that I understand project status from one page. Priority: Should Traceability: FR-15
Acceptance Criteria	• **Given** the project has milestones and tasks, **when** I open the dashboard, **then** I see progress for each milestone, the number of tasks not Done, and tasks due in the next 7 days.

US-40 **View project health**	Actor: Project Manager User Story: As a Project Manager, I want to see overdue tasks, blocked tasks and the recent activity level, so that I can spot problems early. Priority: Should Traceability: FR-22
Acceptance Criteria	• **Given** tasks exist, **when** I open project health, **then** overdue tasks (past due date, not Done) and tasks labelled "Blocked" are listed. • **Given** the project has feed events, **when** I open project health, **then** activity level shows Low (fewer than 5 events in 7 days), Medium (5 to 20) or High (more than 20).

4.8 AI Project Copilot

US-41 **Ask the Copilot a question**	Actor: Project Member User Story: As a Project Member, I want to ask the Copilot a free-text question of up to 500 characters about my project, so that I find information without searching every record. Priority: Must Traceability: FR-16a; NFR-10a, NFR-10b
Acceptance Criteria	• **Given** I am a member of the project, **when** I submit a question, **then** I receive an answer, typically within 10 seconds. • **Given** processing takes longer than 2 seconds, **when** I wait, **then** a loading indicator is shown until the answer or an error arrives.

US-42 **Get answers grounded in project data**	Actor: Project Member User Story: As a Project Member, I want Copilot answers to be based only on project data I am allowed to see, with sources shown, so that I can trust and verify them. Priority: Must Traceability: FR-16b, FR-36; NFR-10b
Acceptance Criteria	• **Given** I ask a question the project data can answer, **when** I receive the answer, **then** it is built only from this project's messages, tasks, decisions and GitHub activity that I can access, and links to the items used. • **Given** a request would exceed the 8,000-token input limit, **when** the request is prepared, **then** the context is trimmed to fit the limit.

US-43 **Be told when data is insufficient**	Actor: Project Member User Story: As a Project Member, I want the Copilot to say so when project data cannot answer my question, so that I am not misled by invented answers. Priority: Must Traceability: FR-16c
Acceptance Criteria	• **Given** the project data does not contain the answer, **when** I ask the question, **then** the Copilot says it does not have enough project data and gives no unsupported project-specific claim.

US-44 **Catch me up**	Actor: Project Member User Story: As a Project Member, I want to request a summary of what happened in a project since my previous login, so that I can quickly understand changes after being away. Priority: Must Traceability: FR-17; NFR-10a, NFR-10b
Acceptance Criteria	• **Given** activity occurred since my previous successful login, **when** I choose Catch Me Up, **then** I receive a summary of that activity using only data I can access, with a loading indicator if it takes over 2 seconds. • **Given** nothing happened since my previous login, **when** I choose Catch Me Up, **then** I am told there is nothing new. • **Given** the AI provider is unavailable, **when** I choose Catch Me Up, **then** I see an "AI temporarily unavailable" message and the rest of FORGE keeps working.

4.9 Discovery and Portfolio

US-45 **Browse and filter public projects**	Actor: Visitor User Story: As a Visitor, I want to browse public projects and filter by domain, skill tag or status, or search by keyword, so that I can find projects that interest me. Priority: Should Traceability: FR-18, FR-30
Acceptance Criteria	• **Given** public and private projects exist, **when** I open Discover, **then** only public projects are listed. • **Given** I apply a domain, skill-tag or status filter, **when** the list refreshes, **then** only matching projects are shown. • **Given** I enter a keyword, **when** I search, **then** projects whose name or description contains it are returned.

US-46 **View a public project page**	Actor: Visitor User Story: As a Visitor, I want to open a public project's read-only page, so that I can understand the project before asking to join. Priority: Should Traceability: FR-20
Acceptance Criteria	• **Given** a project is public, **when** I open its page without being a member, **then** I see its name, description, domain, status and member names, and I cannot edit anything. • **Given** I am not a member, **when** I view the page, **then** channels, messages, tasks and files are not shown.

US-47 **Request to join a public project**	Actor: Registered User User Story: As a Registered User, I want to request to join a public project, so that I can take part in work I am interested in. Priority: Should Traceability: FR-20
Acceptance Criteria	• **Given** I am logged in and not a member, **when** I submit a join request with an optional message, **then** the request is pending and project managers can see it. • **Given** I already have a pending request, **when** I submit another, **then** I am told one is already pending. • **Given** I am not logged in, **when** I choose Request to join, **then** I am asked to log in first.

US-48 **Find a collaborator by skill**	Actor: Registered User User Story: As a Registered User, I want to search users by skill tags, so that I can find people with relevant skills. Priority: Should Traceability: FR-19, FR-29
Acceptance Criteria	• **Given** users have skill tags, **when** I search with one or more tags, **then** users matching at least one tag (case-insensitive) are returned, ordered by number of matching tags. • **Given** a user has opted out of collaborator search, **when** I search, **then** that user is not returned.

US-49 **Generate my portfolio**	Actor: Registered User User Story: As a Registered User, I want to generate a shareable portfolio of my completed projects, my roles and contribution statistics, so that I can present my project experience. Priority: Should Traceability: FR-21; NFR-12
Acceptance Criteria	• **Given** I belong to projects marked Completed, **when** I open my portfolio, **then** each is listed with my role. • **Given** I open my portfolio, **when** statistics are shown, **then** they show tasks completed, decisions recorded and messages posted per project, no more than 24 hours old. • **Given** my portfolio is public, **when** I copy its link, **then** anyone with the link can view it read-only.

4.10 Public Social Feed

US-50 **Create a public post**	Actor: Registered User User Story: As a Registered User, I want to publish a post with text and optionally one image, independent of any project, so that I can share content with the FORGE community. Priority: Must Traceability: FR-23; NFR-16, NFR-26
Acceptance Criteria	• **Given** I enter text of up to 500 characters and optionally one JPEG, PNG or WebP image up to 5 MB, **when** I submit, **then** the post is published and I see a success message. • **Given** I have already created 10 posts in the last hour, **when** I submit another, **then** it is rejected with a rate-limit message. • **Given** the post fails validation, **when** I submit, **then** I see an error explaining why.

US-51 **Edit or delete my post**	Actor: Registered User User Story: As a Registered User, I want to edit or delete my own public post, so that I can correct or remove what I shared. Priority: Should Traceability: New (review); FR-23
Acceptance Criteria	• **Given** I own a post, **when** I edit its text and save, **then** the post shows the new text marked "edited". • **Given** I own a post, **when** I delete it and confirm, **then** it disappears from all feeds and its votes no longer count. • **Given** I do not own the post, **when** I try to edit or delete it, **then** the request is rejected with 403.

US-52 **Vote on a post**	Actor: Registered User User Story: As a Registered User, I want to upvote or downvote a public post once, so that I can express my reaction without duplicate voting. Priority: Must Traceability: FR-24; NFR-13, NFR-14, NFR-16
Acceptance Criteria	• **Given** I have not voted on a post, **when** I upvote or downvote it, **then** the post score changes by one. • **Given** I have already voted on a post, **when** I vote the same way again, **then** the server accepts only one vote, so the score does not change. • **Given** many users vote on the same post at the same moment, **when** the votes are processed, **then** every valid vote is counted exactly once.

US-53 **Browse the trending feed**	Actor: Registered User User Story: As a Registered User, I want to browse a global feed of public posts ranked by trending score, so that I discover content beyond my own projects. Priority: Must Traceability: FR-25; NFR-1a, NFR-15
Acceptance Criteria	• **Given** public posts exist, **when** I open the trending feed, **then** posts are ordered by score = (upvotes - downvotes) / (age in hours + 2)^1.5, highest first. • **Given** votes were cast after the last ranking run, **when** 5 minutes pass, **then** scores are recalculated and the order updates. • **Given** up to 100 users open the feed at once, **when** it loads, **then** it loads within 3 seconds.

US-54 **Follow users and see public counts**	Actor: Registered User User Story: As a Registered User, I want to follow and unfollow other users and see their follower and total-likes counts, so that I can build connections and see public engagement. Priority: Must Traceability: FR-26
Acceptance Criteria	• **Given** I view another user's profile, **when** I choose Follow, **then** I follow them and their follower count increases by one. • **Given** I follow a user, **when** I choose Unfollow, **then** their follower count decreases by one. • **Given** I view any profile, **when** the page loads, **then** it shows the follower count and total likes (total upvotes received across that user's posts).

US-55 **Report a public post**	Actor: Registered User User Story: As a Registered User, I want to report a public post and choose a reason, so that harmful content gets reviewed. Priority: Must Traceability: FR-27
Acceptance Criteria	• **Given** I see a public post, **when** I choose Report, select a reason and submit, **then** the report appears in the moderator queue. • **Given** I have already reported that post, **when** I report it again, **then** I am told it was already reported.

4.11 Moderation and Platform Administration

US-56 **Moderate reported posts**	Actor: Content Moderator User Story: As a Content Moderator, I want to review reported posts and remove them, warn the author or dismiss the report, so that reported content is handled consistently. Priority: Must Traceability: FR-27; NFR-19, NFR-22
Acceptance Criteria	• **Given** reports exist, **when** I open the moderator queue, **then** I see each reported post with its reason and report count. • **Given** I review a report, **when** I choose Remove, Warn or Dismiss, **then** the post is deleted, or the author receives an in-app warning, or the report is closed with no change; the report is marked resolved. • **Given** I am not a moderator, **when** I request the queue or an action, **then** the request is rejected with 403.

US-57 **Assign and revoke moderators**	Actor: Platform Admin User Story: As a Platform Admin, I want to give or take away the Content Moderator role for a user, so that only trusted people moderate content. Priority: Must Traceability: Stakeholder definition; NFR-19, NFR-20
Acceptance Criteria	• **Given** I am a Platform Admin, **when** I grant the moderator role to a user, **then** that user can access the moderator queue on their next request. • **Given** I revoke the role, **when** the user requests the queue, **then** access is refused. • **Given** I am not a Platform Admin, **when** I try to change roles, **then** the request is rejected with 403.

US-58 Suspend and reactivate accounts	Actor: Platform Admin User Story: As a Platform Admin, I want to suspend or reactivate a user account, so that abusive users can be stopped. Priority: Should Traceability: Stakeholder definition; NFR-20, NFR-21, NFR-22
Acceptance Criteria	• **Given** I suspend an account, **when** the user tries to log in or use an existing session, **then** access is refused and active sessions are invalidated. • **Given** I reactivate the account, **when** the user logs in, **then** login succeeds.

5. Non-Functional Requirements

These are system qualities, not features, so they are written as measurable requirements instead of user stories. Source references use the original NFR numbering.

Source ref	Quality	Measurable requirement	Priority
NFR-1a	Feed performance	Project and global feeds load within 3 seconds with up to 100 concurrent users	Must
NFR-1b	Message latency	Channel and direct messages are delivered in under 1 second	Must
NFR-2	Capacity	At least 100 concurrent users supported under class-demo test conditions	Must
NFR-3a	Password storage	Passwords are salted and hashed; never stored or logged in plain text	Must
NFR-3b	Session expiry	JWT expires after 24 hours; expired or logged-out tokens return 401	Must
NFR-3c	Brute-force protection	5 failed logins lock login for 15 minutes	Must
NFR-4	Transport security	All traffic uses HTTPS/TLS	Must
NFR-19 to 22	Access control	Every protected API checks authentication and authorization on the server, using the Section 2 matrix; unauthorized requests are rejected	Must
NFR-5a, NFR-25	Data consistency	Concurrent edits to a task are detected (version check) and never silently overwritten	Must
NFR-13, NFR-14	Vote integrity	One vote per user per post, enforced on the server; vote updates are atomic under simultaneous voting	Must
NFR-21, NFR-23, NFR-26	Validation and feedback	All input is validated on the server; failures return a meaningful message; important actions show visible success or error feedback	Must
NFR-6	Notification latency	Notifications appear within 5 seconds of the triggering event	Must
NFR-5b	Kanban responsiveness	Optimistic UI shows a dropped card in its new column within 300 ms	Should
NFR-7	Responsive layout	Usable at 1280 px (laptop) and 375 px (phone) browser widths with no broken layout or horizontal scrolling	Should
NFR-8	Upload limit	Files over 10 MB are rejected with a clear error	Should
NFR-9, NFR-24	AI isolation	If the LLM provider is unavailable, chat, tasks and feeds keep working; only AI features show an unavailable message	Should
NFR-10a	AI response time	Typical response within 10 seconds; loading indicator shown after 2 seconds	Should
NFR-10b	AI context limit	Each request stays within 8,000 input tokens	Should

Source ref	Quality	Measurable requirement	Priority
NFR-11a	Webhook resilience	Failed GitHub webhook processing is retried 3 times with exponential backoff; the final failure is logged	Should
NFR-11b	GitHub rate limits	FORGE stays within GitHub API rate limits and pauses calls when limits are reached	Should
NFR-12	Portfolio freshness	Contribution statistics are at most 24 hours old	Should
NFR-15	Trending freshness	Trending scores are recalculated at least every 5 minutes	Should
NFR-16	Abuse protection	Per user: 10 posts per hour and 30 votes per minute	Should
NFR-17	Availability	Deployed app is reachable during the demo window; free-tier cold starts show a loading or warm-up message	Should
NFR-27	Consistent interface	The same action uses the same wording and interaction pattern in every module	Should
NFR-28	Loading feedback	Any operation taking over 1 second shows a loading or progress state	Should
NFR-29, NFR-32	Modularity	Major modules are separate; a change in one module does not require changes in others; new features can be added without restructuring	Should
NFR-18, NFR-30, NFR-31	Tests, CI and API docs	New backend routes have automated tests before merge; CI runs the tests; each endpoint's request and response are documented; coding conventions are checked automatically	Should

6. External Dependencies and Constraints

These are not user stories because the actor is a system, a team or an evaluator rather than a user of FORGE.

Item	What FORGE relies on or must respect	Related items
GitHub	Provides OAuth authorization and webhook events for commits, issues and pull requests. Subject to API rate limits.	Section 4.6; NFR-11a, NFR-11b
LLM API provider	Provides language-model processing for the Copilot and Catch Me Up. If unavailable, only AI features are affected.	Section 4.8; NFR-9, NFR-10a, NFR-10b
Vercel, Render, MongoDB Atlas	Host the deployed application and database. Free tiers can cause cold starts and limited capacity.	NFR-17
Development team (10 people)	Works to the agreed module structure and coding conventions, with tests and CI before merge.	NFR-18, NFR-29 to 32
Course instructor / grader	Evaluates the deployed system against the required scope. Individual contributions must be traceable through commit and pull-request history.	Course requirements

7. Traceability Summary

Requirement	User stories
FR-1	US-01, US-02, US-04
FR-2	US-05
FR-3	US-08
FR-4	US-09, US-13, US-14
FR-5	US-16, US-17, US-18

Requirement	User stories
FR-6	US-19
FR-7	US-20
FR-7a	US-21, US-22
FR-7b	US-24
FR-7c	US-25, US-26
FR-8	US-27, US-28
FR-9	US-29
FR-10	US-18, US-22, US-23, US-30
FR-11	US-31, US-32
FR-12	US-34
FR-13	US-35
FR-14a	US-37
FR-14b	US-38
FR-15	US-23, US-39
FR-16a	US-41
FR-16b	US-42
FR-16c	US-43
FR-17	US-44
FR-18	US-08, US-45
FR-19	US-06, US-07, US-48
FR-20	US-12, US-46, US-47
FR-21	US-07, US-15, US-49
FR-22	US-40
FR-23	US-50, US-51
FR-24	US-52
FR-25	US-53
FR-26	US-06, US-54
FR-27	US-55, US-56
FR-28	US-02, US-03
FR-29	US-05, US-48
FR-30	US-45
FR-31	US-10, US-11
FR-32	US-24
FR-33	US-30
FR-34	US-36
FR-35	US-32, US-33

Requirement	User stories
FR-36	US-42

Stories without an FR reference (moderator assignment and account suspension) come from the Platform Admin stakeholder definition and NFR-19 to 22.

8. Assumptions and Open Questions

- All proposed values in Section 3 need team confirmation, especially the trending formula, which the source requires to be fixed before implementation.
- Password reset assumes an email-sending service is available in the deployed stack.
- The project domain list used for filtering (US-08, US-45) still needs to be defined.
- Threads (priority Should) were split from channel messages (Must). Confirm this against the source priority for FR-5.
- Catch Me Up is offered to every Project Member, not only Contributors. Confirm.
- "Manage platform configuration" was removed from the Platform Admin story because it could not be tested. Add it back only with concrete actions.
- The new stories marked "New (review)" go beyond the original requirement list. Confirm with the team that they are in scope, or lower them to Won't.

9. How the Review Feedback Was Applied

Review point	Change made
Five items were not user stories	Moved to Section 6 (GitHub, LLM provider, hosting, development team, instructor)
Grammar	All stories now read "I want to ..."
Prompt-leak text and draft wording	All file-citation text and "consolidated version" wording removed
Weak and undefined acceptance criteria	Rewritten as Given/When/Then; undefined terms replaced by values in Section 3
Stories too big	Channels split into 3 stories; Copilot split into 3 (plus Catch Me Up); GitHub split into 3 (plus feed events)
Platform Admin untestable	Replaced with two concrete stories: assign/revoke moderators, suspend/reactivate accounts
Actor confusion; no permission matrix	Actor glossary (Section 1) and role-permission matrix (Section 2) added
Invalid priorities	MoSCoW only (Must, Should, Could)
Missing stories	Added: comments, privacy settings, accept/decline invites, approve join requests, mark project complete, due dates, edit/delete posts, remove/leave project, view profiles, password reset